

Diffusion Language Models: Foundations for Algorithm Design

Working Group Notes¹

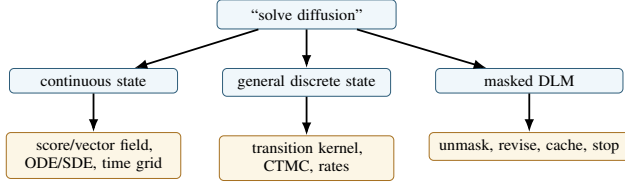


Figure 1. Three distinct meanings of solving diffusion.

Abstract

Diffusion language modeling is not one algorithmic problem. Continuous diffusion, flow matching, general discrete diffusion, and absorbing-mask DLMs change different mathematical objects. This note keeps only the foundations needed for team discussion: the process, learned target, executable sampler, optimization surface, and falsification contract.

1. First-Principles Object

The phrase “solve the diffusion process” is ambiguous in language modeling: it may mean numerical integration in continuous space, reverse simulation of a discrete Markov process, path design and vector-field learning, or control of a masked-DLM sampler. Before discussing algorithms, name four objects:

state, learned conditional, action, budget.

2. Transport Foundations

2.1. Continuous reference point

For continuous data, a common variance-preserving marginal is

$$z_t = \sqrt{\alpha_t}x + \sqrt{1 - \alpha_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (1)$$

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where the schedule moves clean data toward a tractable prior. A learned score induces a reverse SDE or probability-flow ODE; inference chooses its solver and time grid [18, 22]. Continuous text diffusion applies this geometry to embeddings or latent vectors [19].

2.2. Flow matching

Flow matching (FM) learns the velocity of a chosen probability path rather than a score. If a conditional path has velocity $u_t(x_t | x_1)$, the simulation-free objective is

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, x_1, x_t} \|v_\theta(x_t, t) - u_t(x_t | x_1)\|_2^2. \quad (2)$$

Sampling integrates $\dot{z}_t = v_\theta(z_t, t)$. Diffusion paths are a special case, whereas optimal-transport paths need not be diffusions [20].

Discrete Flow Matching (DFM) instead defines probability paths and velocities over categorical states, making the coupling, scheduler, and CTMC sampler explicit [10, 28]. Dirichlet and Fisher flow matching provide alternative simplex or statistical-manifold geometries for discrete sequences [29, 8]. Thus flow matching changes the path and training target; it does not eliminate the inference-time discretization problem.

2.3. Discrete Markov diffusion

Let a token be a one-hot vector in a vocabulary $\mathcal{V}$. D3PM defines a forward Markov chain using categorical transition matrices Q_t :

$$q(z_t | z_{t-1}) = \text{Cat}(z_t; Q_t z_{t-1}), \quad (3)$$

$$q(z_t | x) = \text{Cat}(z_t; \bar{Q}_t x), \quad \bar{Q}_t = Q_t Q_{t-1} \cdots Q_1. \quad (4)$$

The known forward process gives $q(z_{t-1} | z_t, x)$. A neural model replaces the unknown clean x at generation time. Its NELBO contains

$$\mathcal{L}_{\text{diff}} = \sum_t \mathbb{E}_{q(z_t|x)} D_{\text{KL}}[q(z_{t-1} | z_t, x) \| p_\theta(z_{t-1} | z_t)]. \quad (5)$$

D3PM permits uniform, structured, or absorbing transitions [3]. Continuous-time variants use a CTMC rate matrix; SEDD estimates ratios needed by the reverse process [4, 21].

3. Absorbing-Mask Language Diffusion

3.1. Forward process

Introduce a special absorbing mask state $m = [\text{M}]$. For a clean token x , masked diffusion uses

$$q(z_t | x) = \text{Cat}(z_t; \alpha_t x + (1 - \alpha_t)m), \quad (6)$$

with $\alpha_0 \approx 1$, $\alpha_1 \approx 0$, and α_t decreasing. A token either remains clean or jumps to the mask; after it is masked in the forward process, it stays masked. The process is applied independently to response positions, while conditioning or prompt tokens remain visible.

For $0 \leq s < t \leq 1$, define the categorical parameter

$$r_{s,t}(x) = \frac{(1 - \alpha_s)m + (\alpha_s - \alpha_t)x}{1 - \alpha_t}. \quad (7)$$

The posterior has a simple interpretation. If $z_t \neq m$, then the earlier state must equal z_t . If $z_t = m$, the earlier state is either still masked or equal to the clean token:

$$q(z_s | z_t, x) = \begin{cases} \delta_{z_t}, & z_t \neq m, \\ \text{Cat}(z_s; r_{s,t}(x)), & z_t = m. \end{cases} \quad (8)$$

Figure 2 separates one-shot training corruption from the iterative generation loop.

3.2. Reverse parameterization and training

Because clean x is unknown during generation, a denoiser $x_\theta(z_t, t)$ predicts a distribution over clean tokens. Substituting this prediction into Equation 8 defines $p_\theta(z_s | z_t)$. MDLM additionally enforces zero probability for the mask as a clean token and carries already unmasked tokens forward. These choices simplify the continuous-time NELBO to

$$\mathcal{L}_{\text{MDLM}} = \mathbb{E}_{x,t,z_t} \left[\frac{\alpha'_t}{1 - \alpha_t} \log \langle x_\theta(z_t, t), x \rangle \right]. \quad (9)$$

Since $\alpha'_t < 0$, minimizing this expression is a weighted masked-token cross-entropy objective. This connection explains why BERT-like bidirectional Transformers can parameterize generative masked diffusion, but it does not make an ordinary masked language model a DLM automatically [17, 32, 26].

4. Sampler as the Optimization Object

The denoiser returns conditional token distributions; the sampler turns them into state changes. The lifecycle in Figure 2 is therefore the operational algorithm: observe the current canvas, run the denoiser, choose which tokens to commit or revise, update state, and stop under a budget.

For a masked DLM, a compact post-forward observation is

$$s_b = (x_b, M_b, P_b, h_b, c_b, b), \quad (10)$$

where M_b is the mask set, P_b the current model output, h_b trajectory history, c_b cache/system state, and b remaining budget. An action can be represented as

$$a_b = (U_b, R_b, g_b, \text{stop}), \quad (11)$$

where U_b commits positions, R_b revises or verifies positions, and g_b selects a compute mode. This is a history-augmented control problem, not automatically a small MDP: token actions are combinatorial and final quality is observed late. The minimal matched-budget comparison is

$$\max_{\pi} \mathbb{E}_{q \sim \mathcal{D}}[Q_{\pi}(q)] \quad \text{s.t.} \quad \sum_b \text{NFE}_b \leq B. \quad (12)$$

Wall-clock claims additionally require synchronized p50/p95 latency, memory, batching, and policy overhead [14].

Table 1. Where an algorithm can act.

Layer	Decision variable	Falsifying question
Path	transition/rate, mask schedule	Is the gain just a different training target?
Model	backbone, time conditioning	Does it hold at matched parameters/training?
Sampler	time grid, update set, revision	Does it beat matched NFE and wall-clock cost?
Length	canvas, EOS, block factorization	Are failures hidden as unfinished masks?
System	cache, batching, kernels	Are states still valid after revisions?

Three pitfalls usually decide whether an idea is real. A fixed canvas needs a length mechanism [26, 2, 6]; independent token marginals do not guarantee jointly safe parallel commits; and remasking changes both the transition rule and cache validity.

5. Historical and Model Map

Iterative masked decoding predates modern DLMs; continuous diffusion, discrete diffusion, and flow matching developed as related but distinct paths. Figure 3 is a cited reading map, not a lineage claim [31].

Model families differ by probability path, representation, backbone, factorization, length handling, and inference policy. Mamba and Mamba-2 are backbone choices rather than diffusion solvers [13, 7]. The large-model branch is currently represented by LLaDA, Block Diffusion, and Dream [23, 2, 30]; newer work studies trajectory mismatch, masked pretraining, and scaling [16, 24, 27].

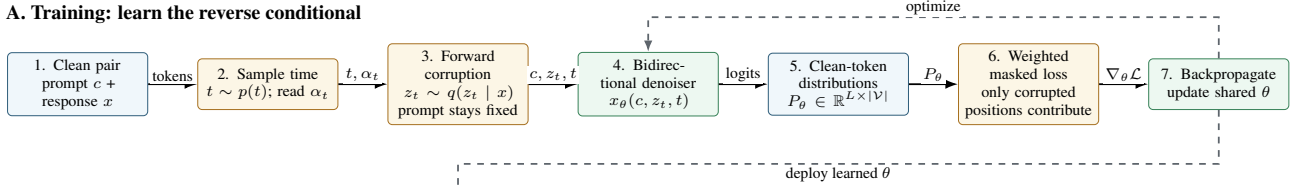
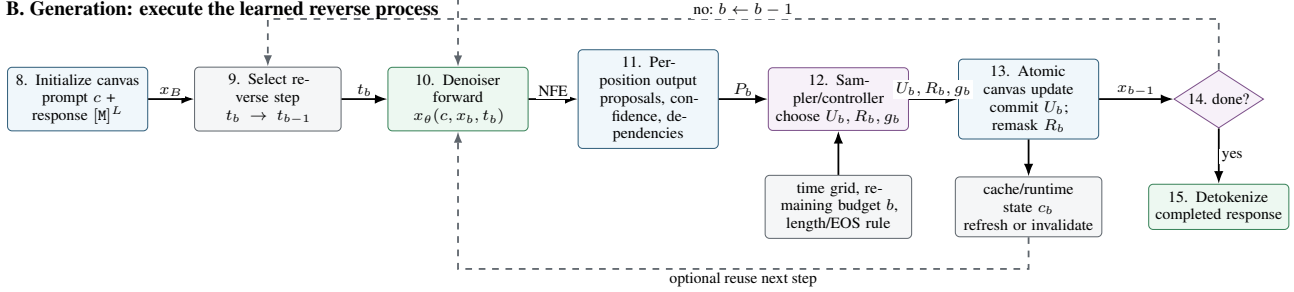
A. Training: learn the reverse conditional

B. Generation: execute the learned reverse process


Figure 2. Masked-DLM lifecycle. Training corrupts once; generation iterates denoising and canvas updates. Colors distinguish data (blue), known operations (orange), model/output (green), control (violet), and state (gray).

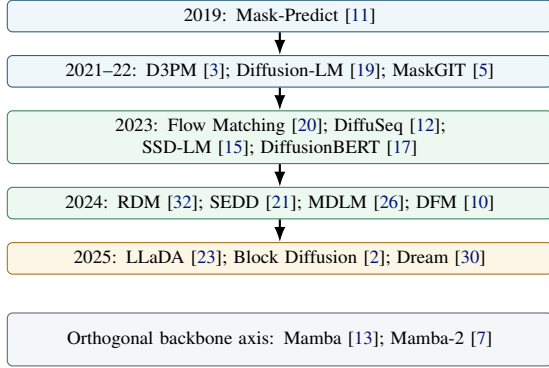


Figure 3. A compact reading history. Each named work has a numeric citation; arrows order process-level foundations, while the gray box is a separate backbone axis.

Table 2. Minimum evidence for a DLM comparison.

Dimension	Report
Model	checkpoint, training changes, tokenizer, time conditioning
Work	reverse steps, total model evaluations, extra scorers/branches
Latency	synchronized end-to-end p50/p95 and policy overhead
Memory	peak allocation and cache footprint
Length	prompt, canvas, generated length, EOS/completion rule
Quality	task metric, likelihood metric if relevant, completion rate
Robustness	second task, length regime, and preferably second model family
Controls	matched-cost baseline and signal-destroying ablation

6. Research Constraints and Evidence

Four constraints should stay visible during brainstorming. First, metric choice changes apparent speedup: few steps can look strong under perplexity while sequence error grows with length [9]. Second, masking has process structure tied to known jump timing [1]. Third, a CTMC solver is not a token scheduler: solvers improve simulation, whereas policies choose token actions [25]. Finally, training and inference paths can mismatch; random masks may not match generation canvases [16].

An experiment must report the quality–cost frontier, not a single operating point. The minimum evidence is compact:

Common failure modes are unfinished masks, length ambiguity, confidence drift across mask ratios, correlated token errors, stale caches, and gains that vanish under matched compute.

7. Research Gate

We change *[layer/action]* using *[signal]* under *[budget]*, because the matched baseline fails on *[diagnostic]*. The idea is falsified if *[matched-cost result]*. If these fields cannot be filled, keep measuring or reading.

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